Quadratic Graphs

Exam code: 0607

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Written by: Naomi C
Reviewed by: Dan Finlay
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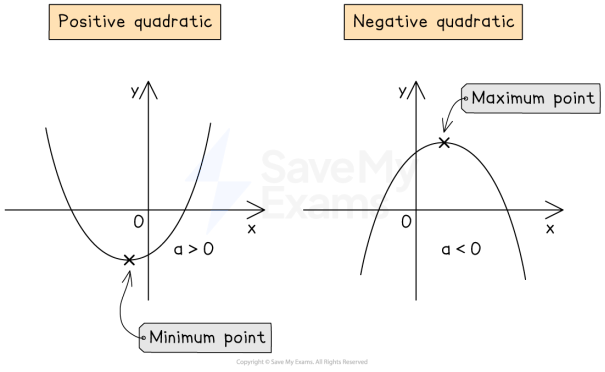
Quadratic Graphs

What is a quadratic graph?

- A **quadratic graph** has the form $y = ax^2 + bx + c$
  - where a is not zero

What does a quadratic graph look like?

- A quadratic graph is a smooth curve with a **vertical line of symmetry**
  - A **positive** number in front of x^2 gives a **u-shaped curve**
  - A **negative** number in front of x^2 gives an **n-shaped curve**
- The shape made by a quadratic graph is known as a **parabola**
- A quadratic graph will **always** cross the y**-axis**
- A quadratic graph crosses the x**-axis twice, once, or not at all**
  - The points where the graph crosses the x-axis are called the **roots**
- If the graph is a **u-shape**, it has a **minimum point**
- If the graph is an **n-shape**, it has a **maximum point**
- Minimum and maximum points are both examples of **a vertex**



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- A simple drawing showing the **key features** is often sufficient
- For a more accurate graph, create a table of values and plot the points

- To **sketch a quadratic graph**:
  - First sketch the x and y-axes
  - Identify the y-intercept and mark it on the y-axis
    - The y-intercept of $y = ax^2 + bx + c$ will be $(0, c)$
    - It can also be found by substituting in $x = 0$
  - Find all root(s) (0, 1 or 2) of the equation and mark them on the x-axis
    - The roots will be the solutions to $y = 0$; $ax^2 + bx + c = 0$
    - You can find the solutions by factorising or using the quadratic formula
  - Identify whether the number a in $ax^2 + bx + c$ is positive or negative
    - A positive value will result in a u-shape
    - A negative value will result in an n-shape
  - Sketch a smooth curve through the x and y-intercepts
    - Mark on any axes intercepts
    - Mark on the coordinates of the maximum/minimum point if you know it

Worked Example

Sketch the graph of $y = x^2 - 5x + 6$ showing the x and y intercepts clearly.

Solution:

The $+c$ at the end is the y-intercept

$$y\text{-intercept: } (0, 6)$$

Factorise the quadratic expression

$$y = (x - 2)(x - 3)$$

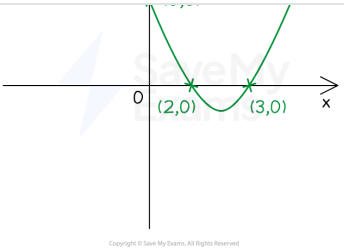
Solve $y = 0$

$$(x - 2)(x - 3) = 0, \text{ so } x = 2 \text{ or } x = 3$$

So the x-intercepts are given by the coordinates

$$(2, 0) \text{ and } (3, 0)$$

It is a positive quadratic graph, so will be a u-shape



How do I find the coordinates of the vertex?

- For a quadratic graph written in the form $y = a(x-p)^2 + q$
  - the minimum or maximum point has **coordinates** (p, q)
- Beware: there is a **sign change** for the x-coordinate
  - A curve with equation $y = (x-3)^2 + 2$, has a minimum point at $(3, 2)$
  - A curve with equation $y = (x+3)^2 + 2$, has a minimum point at $(-3, 2)$
- The value of a does **not affect the coordinates** of the vertex but it **will change the shape** of the graph
  - If it is **positive**, $a > 0$, the graph will be a u-shape
    - The curve $y = 5(x-3)^2 + 2$ has a **minimum** point at $(3, 2)$
  - If it is **negative**, $a < 0$, the graph will be an n-shape
    - The curve $y = -8(x-3)^2 + 2$ has a **maximum** point at $(3, 2)$

Worked Example

Sketch the graph of $y = (x-3)^2 + 4$ showing the y-intercept and the coordinates of the vertex.

Answer:

It is a positive quadratic, so will be a u-shape
The vertex will therefore be a minimum

The vertex of $y = a(x-p)^2 + q$ has coordinates (p, q)
The minimum point is therefore

$$(3, 4)$$

To find the y-intercept, substitute in $x = 0$

$$y = (0-3)^2 + 4$$
$$y = 9 + 4$$
$$y = 13$$

y-intercept: $(0, 13)$

How do I find the equation of a quadratic from its graph?

- If the **vertex** and **one other point** are known
  - Use the form $y = a(x - p)^2 + q$ to fill in p and q
    - The vertex is at (p, q)
  - Then substitute in the other known point (x, y) to find a
- If the **roots** (x-intercepts) and **one other point** are known
  - Use the form $y = a(x - x_1)(x - x_2)$ to fill in x_1 and x_2
    - The roots are at $(x_1, 0)$ and $(x_2, 0)$
  - Then substitute in the other known point (x, y) to find a
- If $a = 1$ then you only need either the **vertex** or the **roots**

Worked Example

(a) Find the equation of the graph below.

Answer:

The graph shows the roots and a point on the curve (in this case the y-intercept)

The roots are at $(x_1, 0)$ and $(x_2, 0)$

$$y = a(x-2)(x-3)$$

Substitute in the other known point (0, 24) to find a

$$\begin{aligned} 24 &= a(0-2)(0-3) \\ 24 &= a(-2)(-3) \\ 24 &= 6a \\ 4 &= a \end{aligned}$$

Write the full equation

$$\mathbf{y = 4(x-2)(x-3)}$$

You could also write this in expanded form: $y = 4x^2 - 20x + 24$

(b) Find the equation of the graph below.

Answer:

The graph shows the vertex and a point on the curve

Use the form $y = a(x-p)^2 + q$ to fill in p and q by inspection

The vertex is at (p, q)

$$y = a(x-9)^2 - 16$$

Substitute in the other known point (2, 82) to find a

$$\begin{aligned} 82 &= a(2-9)^2 - 16 \\ 82 &= a(-7)^2 - 16 \\ 82 &= 49a - 16 \\ 98 &= 49a \\ 2 &= a \end{aligned}$$

Write the full equation

$$\mathbf{y = 2(x-9)^2 - 16}$$

You could also write this in expanded form:
$y = 2x^2 - 36x + 146$

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